

Definitive Technical Approach

Satellite-Based Ground Deformation Monitoring, Chimanimani District, Zimbabwe

This document fixes the final technical approach for the project: the exact pipeline stages, the tools and data sources used at each stage, the mathematics governing each step, and the literature each decision is grounded in. It is meant to be the single reference the whole group works from from this point forward.

1 Study Area

Chimanimani District, Eastern Highlands, Zimbabwe. Selected on the basis of documented landslide activity during Cyclone Idai (March 2019), which provides both a real hazard justification and an existing before/after satellite reference for validation. Exact bounding coordinates to be finalised in Phase 0 using the known landslide-affected zones reported in post-Idai satellite analysis (see Section 6, ref. 8).

2 Pipeline Overview

Stage 0 — Area definition. Fix the AOI bounding box and pull a reference DEM.
Stage 1 — Data acquisition. Download a Sentinel-1 SLC stack covering the AOI.
Stage 2 — InSAR processing. Generate interferograms, unwrap phase, invert via SBAS to get a displacement time series.
Stage 3 — Anomaly detection. Flag pixels/points whose deformation deviates from their own baseline trend.
Stage 4 — Validation. Cross-check flagged anomalies against known Idai landslide scars and optical imagery.

3 Stage 0: Study Area and Elevation Data

Tool/source: Copernicus GLO-30 DEM (30 m resolution, global coverage, free).

Access: OpenTopography or directly via Google Earth Engine (`COPERNICUS/DEM/GLO30`).

Purpose: required to correct the topographic phase term $\delta\phi_{\text{topo}}$ in Stage 2.

4 Stage 1: Data Acquisition

Source: Sentinel-1 SLC (Single Look Complex) imagery, C-band, 5.55 cm wavelength.

Access points: Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu>) or Alaska Satellite Facility Vertex (<https://search.asf.alaska.edu>), both free.

Requirement: minimum 15–20 acquisitions, same track/orbit direction, spanning at least 12 months to sample seasonal atmospheric variation and, ideally, bracketing the Idai event window (March 2019) if historical archive coverage allows a retrospective comparison.

5 Stage 2: InSAR Processing (SBAS)

Tool: MintPy (open-source Python package), following the SBAS approach of Berardino et al. (2002) and Lanari et al. (2007). Interferogram generation itself is typically handled upstream

via ISCE2 or HyP3 (ASF’s on-demand processing service) before MintPy performs the time-series inversion.

5.1 Interferometric phase model

$$\delta\phi = \frac{4\pi}{\lambda} \Delta d_{LOS} + \delta\phi_{\text{topo}} + \delta\phi_{\text{atm}} + \delta\phi_{\text{noise}}$$

5.2 Phase unwrapping

Minimum cost flow unwrapping (standard in MintPy/SNAPHU) recovers continuous phase from the wrapped (mod 2π) measurement.

5.3 Time-series inversion

For N acquisitions and M small-baseline interferometric pairs:

$$\mathbf{A}\mathbf{v} = \frac{\delta\phi}{4\pi/\lambda}, \quad \hat{\mathbf{v}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \left(\frac{\delta\phi}{4\pi/\lambda} \right)$$

Cumulative displacement is obtained by integrating $\hat{\mathbf{v}}$ over time, per pixel, giving the full deformation time series that feeds Stage 3.

6 Stage 3: Anomaly Detection

Baseline method (primary deliverable): per-pixel trend fit $\hat{d}_i(t)$, flag where

$$|d_i(t) - \hat{d}_i(t)| > k\sigma_i, \quad k \approx 2-3$$

Extension (if time permits): autoencoder trained on normal deformation patterns; reconstruction error above threshold treated as anomaly score, following the general susceptibility-modelling approach used in the Quetta, Pakistan study (Section 6, ref. 5).

7 Stage 4: Validation

Cross-reference flagged points against Sentinel-2 optical imagery (before/after Idai) and NASA Earth Observatory’s documented landslide-affected zones (Section 6, ref. 8) to check whether known landslide locations correspond to elevated pre-event deformation signal or post-event ongoing instability.

8 Scope Boundary: What This Project Does Not Claim

This pipeline is retrospective by nature (Sentinel-1 revisit cycle of 6–12 days plus processing time) and cannot support real-time emergency response. It is a screening and early-warning tool. Real-time active-duty monitoring, should it be pursued beyond this project, would require Ground-Based InSAR, continuous GNSS, or IoT tiltmeter networks deployed at specific sites flagged by this pipeline — a distinct future-work direction, not a feature of the current build.

9 Literature

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